

Correction to xtea

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We give below a response to the Newsgroup note on Block TEA

The newsgroup sci.crypt.research published an effective attack on the Block TEA code by Markku-Juhani Saarinen. We agree the attack is valid and usage where many undetected ciphertext attacks can be made should be rejected. The attack is effective against Block TEA, not the original TEA or XTEA.

We have considered the lack of back propagation that was pointed out and it seems easy to make the decoding difference propagation about as fast as the forward coding mode. A simple and apparently safe way is detailed below. The shifting values are different so that $n=2$ will not give problems. The rearrangement makes the plus and XOR operations alternate to slightly increase non linearity. The simple solution is about 30% may be better. It has the same simple interface as before.

Basically $v[m] += f(v[m-1])$ is changed to $v[m] += f(v[m-1], v[m+1])$ where m increases or decreases modulo n .

However, the re-arrangement may have introduced extra problems and we would be pleased to hear if any are detected.

Provisional routine.

```
#define MX (z>>5^y<<2)+(y>>3^z<<4)^(sum^y)+(k[p&3^e]^z) ;
```

```
long btea( long * v, long n , long * k ) {  
    unsigned long z=v[n-1], y=v[0], sum=0,e,  
        DELTA=0x9e3779b9 ;
```

```

long m, p, q ;
if ( n>1) {
    /* Coding Part */
    q = 6+52/n ;
    while ( q-- > 0 ) {
        sum += DELTA ;
        e = sum >> 2&3 ;
        for ( p = 0 ; p < n-1 ; p++ )
            y = v[p+1],
            z = v[p] += MX
        y = v[0] ;
        z = v[n-1] += MX
    }

    return 0 ; }

    /* Decoding Part */
else if ( n <-1 ) {
    n = -n ;
    q = 6+52/n ;
    sum = q*DELTA ;
    while (sum != 0) {
        e = sum>>2 & 3 ;
        for (p = n-1 ; p > 0 ; p-- )
            z = v[p-1],
            y = v[p] -= MX
        z = v[n-1] ;
        y = v[0] -= MX
        sum -= DELTA ; }
    return 0 ; }
return 1 ; } /* Signal n=0,1,-1 */

```

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